Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Python / LangGraph / React · Healthcare AI & HIPAA Compliance · HITL & Agentic Systems · Eval Infrastructure

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SUMMARY

AI Engineer with 9 years of production software ownership. Built LeaseI -- an 8-agent LangGraph system with HITL guardrails on regulated decisions: the LLM extracts and reasons; a human or deterministic logic makes every final regulated call. Core design enforced in code: eligibility questions never resolved by AI. 93.6% precision across 1,240 test cases. Built AgentProbe -- an open-source eval platform: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, and pass@k harness -- the infrastructure that proves agents work before they touch real users. Healthcare AI in production: AI on FHIR (HIPAA-compliant PHI, FHIR APIs, RBAC) and Entrata PHI compliance (AWS S3 migration, zero security incidents across 7 years). At Magizhs Tech Solution, designed and delivered a production LangGraph system for an external client end-to-end. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & Agentic Systems: Python, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Large Language Models (LLMs), Generative AI, HITL (Human-in-the-Loop), Guardrails, Output Validation, Hallucination Detection, RAG Pipelines, ChromaDB, Embeddings, Prompt Engineering, Structured Outputs, LLM Evaluation, LLM-as-Judge, Agent Tracing, Eval Infrastructure, Pass@k, Regression Gate, LLMOps, HIPAA Compliance, PHI Data Handling, RBAC, Data Privacy

Full-Stack & Backend: Python, JavaScript, TypeScript, Java, SQL, React.js, Angular, FastAPI, Flask, REST APIs, Microservices, RabbitMQ, Redis, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue, Lambda), Azure, GCP

Data & ML: PostgreSQL, MySQL, Oracle, PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, Fine-tuning, NLP, BERT, spaCy, OCR, Document Intelligence, ETL, Data Pipelines, Git, Linux, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI Engineer (Client Delivery)

- Designed and delivered a production LangGraph agentic system for an external client -- owned the full technical roadmap from architecture through deployment: LLM selection, RAG layer, prompt-engineered guardrails, output validation, and integration with enterprise data sources.
- Shipped a working system in weeks, not months -- built the eval harness alongside the product so every change was validated before it reached the client's data.

Entrata (PropTech SaaS · 100K+ MAU · HIPAA/PHI · regulated production systems)

Sep 2018 - Aug 2024

Senior Software Engineer

- **HIPAA and PHI compliance in production:** Led AWS S3 document migration with Libsodium-encrypted pipelines, 100% PHI compliance, zero security incidents across the entire migration and 7 years of ongoing data ownership.
- Owned compliance-critical data systems (180,000 HUD records, housing eligibility for 100K+ residents) -- zero violations in 7 years; validation, reconciliation, and rollback on every release cycle.
- Shipped 20+ full-stack production modules (Python, JavaScript/TypeScript, React, PostgreSQL, RabbitMQ) across 120+ Agile sprints; led Docker microservices redesign (50% scalability improvement); eliminated 80% of manual workload through Python/Bash automation.
- Optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers; conducted 300+ peer code reviews; built internal debugging tooling that cut investigation time by 40%.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Fine-tuned a custom classification model architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented evaluation results to faculty.
- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest anomaly detection across 3 research pipelines.

PROJECTS

AgentProbe: AI Eval Infrastructure & LLMOps Platform

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing

- Open-source eval platform: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, pass@k harness, agent tracing, and multi-provider LLM routing -- the infrastructure that tells the difference between a good agent output and a confident wrong answer, before it reaches production.

LeaseI: 8-Agent HITL System with Document Intelligence & RAG

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Production agentic system with a core rule enforced in code: the LLM extracts and reasons; a human or deterministic logic makes every final regulated call. HITL escalation when confidence is too low; eligibility questions always route to a human, never auto-resolved. OCR ingestion, section-aware chunking, structured data extraction from 1,240 PDFs -- 93.6% precision, 87.9% F1.

AI on FHIR: HIPAA-Compliant Healthcare AI System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR APIs · spaCy · BERT · Docker · HIPAA · RBAC

- HIPAA-compliant AI system over structured PHI -- RBAC access controls, NLP-to-FHIR API translation (EN/ES), LLM + BERT + embeddings integration; data privacy enforced at architecture level. Reduced manual clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017